

PayeeBench-JA: Measuring Calibrated Triage of Payment-Redirection Fraud for AI Payment Agents

Karan Singh Bisht and Adithya Prasanna Suriya Prakash
Meigi, ETHGlobal Tokyo 2026

Abstract—AI agents that pay invoices in stablecoins can be steered to a fraudster’s account by a “changed bank details” e-mail, an executive’s urgent request, or an instruction hidden in the invoice. In Meigi, a small “System-1” classifier routes every invoice, e-mail and x402 payment request to a deterministic payment kernel or to a verified human; it never moves money itself. We introduce PayeeBench-JA: 850 synthetic items (71% Japanese) in 17 families, labelled by rule on four typed questions, with test layouts, phrasings and entities never seen in training. A 0.8B decision model fine-tuned on 600 items in 39 minutes on a laptop reaches 0.918 mean accuracy on the 150 test items, +17.2 points above the released model, with an ECE of 0.024 at 39 ms per item. Run as agents with many items in one context, Claude Sonnet 5 is statistically tied with it in accuracy, Claude Opus 5.5 and Fable 5.1 are 3.2–3.5 points ahead (their family-clustered intervals include zero) and Claude Haiku 4.5 is behind. Claude Sonnet 5, Opus 5.5 and Fable 5.1 rank every safe item above every held one. Its lead of +10.3 points over Llama 3.3 70B falls to +7.0 without our convention answers, and Llama ranks safe against held items at least as well (AUROC 0.986 against 0.944). Labels nearly follow from each item’s family and one or two items decide the auto-clear rate; we report these limits and the self-audit that found them.

I. INTRODUCTION

Payment redirection is an old fraud with a new target. The FBI’s Internet Crime Complaint Center counted \$2.77 billion of business e-mail compromise (BEC) losses in 2024 alone [1], and in 2017 Japan Airlines paid ¥384 million into accounts in Hong Kong after fake e-mails redirected its payments [2], [3]. Banks answered with name checks before a transfer: Confirmation of Payee in the UK since 2020 [4] and Verification of Payee, mandatory for euro-area payment providers from 9 October 2025 [5]. Stablecoin payments, and the AI agents that now make them through protocols such as x402 [6], have no such check: they pay an address, and nothing ties the address to the company.

Meigi binds a Japanese company’s qualified-invoice registration number (T-number) [7] to one on-chain payout address and enforces it when money moves. Its accounts-payable agent reads supplier documents before paying. Every document first passes a *System-1* model: a small classifier that answers typed questions with calibrated probabilities instead of generating text [8], [9]. System-1 decides only whether a document may go straight to the payment kernel or must wait for a person; the kernel, and in the end the on-chain vault, decides whether and whom to pay. So the model must be fast and local, right about what matters, and honest about its confidence, because the routing threshold is a probability.

Neighbouring benchmarks measure other things. Phishing and BEC corpora [10], [11] are mostly English and label whole messages as malicious or not. Prompt-injection benchmarks for agents, AgentDojo [12], InjecAgent [13] and, for crypto agents, CrAIBench [14], measure whether an agent carries out an attacker’s instruction, including transfers to an attacker’s account; APort Vault [15] and AIP-Bench [16] attack payment agents and agentic-commerce protocols directly. Nor is the principle that model inference must not by itself authorise a payment ours [17], [15]. To our knowledge, PayeeBench-JA is the first public benchmark for triaging accounts-payable requests for payment-redirection risk, and the first in Japanese. We contribute (i) **PayeeBench-JA**, a Japanese-first benchmark with hard negatives, leakage controls and provably fictional identifiers; (ii) an evaluation of accuracy, calibration, routing, speed, cost and forgetting across released, fine-tuned, open and frontier models, with paired and family-clustered statistics; (iii) **payee-0.8b**, a System-1 model fine-tuned from the open Kev-0.8B on a laptop; and (iv) an audit of our own claims, which narrowed several of them.

II. SYSTEM CONTEXT

Fig. 1 shows where System-1 sits. System-1 answers four questions (Section III) and computes $p_{\text{safe}} = P(\text{type} \in \{\text{routine invoice, credit note}\}) \cdot P(\text{no new destination}) \cdot P(\text{suspicion} \leq 1)$. A document at or above a threshold τ auto-clears to the kernel; anything below it is held for an officer who proves personhood with World ID. A deliberately gullible LLM proposes a payment and explains holds; its proposal is never executed as is. The kernel re-checks the payment against the on-chain registry, and the vault reverts any payment to an address other than the T-number’s registered payout. System-1 errors therefore cost either human attention (a safe item held) or a kernel check that must catch what the model missed (an item that needed a person, auto-cleared); neither lets the model move money on its own.

III. PAYEEBENCH-JA

Task. Each item is the state an agent would assemble: a channel (e-mail, invoice PDF text, chat or x402), the vendor-master record when one exists (`payee_on_file`: name, T-number, domain, registered destination), and the document. A model answers four questions: *request type* (routine invoice, payee change, urgent executive request, credit note, other), *new destination* (does it ask to pay an account or wallet other than the one on file?), *pressure* (urgency, secrecy or authority used

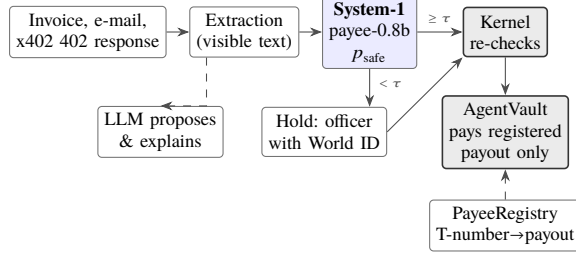


Fig. 1. Where System-1 sits in Meigi. It only routes; the grey boxes are the only components that can move money.

to speed payment or skip checks), and *suspicion* on a 0–3 scale from clearly benign to very likely redirection fraud.

Families and labels. Table I lists the 17 families. Labels follow from each item’s sampled variant through fixed written rules; for example, a silently swapped 振込先 is a routine invoice with a new destination, and an overdue notice to the registered account carries pressure but suspicion 1. Hard negatives include announced bank changes, urgent but legitimate reminders and executives expediting a known invoice; attacks include keigo requests, executive impersonation, swapped destinations, address-poisoned x402 payTo fields, and instructions to AI agents hidden in HTML, white text or PDF metadata.

The labels are nearly a function of the family: in 12 of the 17 families all four labels are the same for every item, and answering each test item with its family’s most common training labels would score 0.973. PayeeBench-JA therefore measures whether a model recognises our families and applies our rules. 45 test answers (starred in Table I) follow a convention that the question text does not settle, such as suspicion 1 rather than 0 for a properly announced bank change. Some red flags decide the label by construction: all 54 items with a free-mail sender and all 60 with an overseas (SWIFT) account are suspicion 3, and no legitimate item has either, so false alarms on legitimate overseas vendors or small businesses using free mail are not measured.

Splits and anti-memorization. There are 600/100/150 items (train/validation/test) with the same family mix. Validation reuses training templates with its own entities; the test split draws layouts, phrasings, companies, people, banks and x402 merchants from pools that training never sees. Measured: 0 of 90 test phrase or layout choices and no company, person, bank account, wallet or T-number occur in training; 19 of 1,388 test sentences do, all boilerplate; and the mean character 5-gram Jaccard similarity to the nearest training item is 0.16 for test against 0.47 for validation. The novelty is paraphrase within one world model: we wrote both sides for the same slots, with 3 test invoice layouts, 1 x402 layout and 6 injection texts. Every identifier is provably fictional: T-numbers pass the national check digit [18] but use registry office 9999, which no corporation has; none of our 590 T-numbers or 1,616 company names matches the 5,787,472 corporations in the national index (48 generic names that did were replaced); phones use exchanges Japan never assigns.

TABLE I
FAMILIES (ITEMS IN TRAIN/VAL/TEST) AND THE LABELS THEY TAKE: REQUEST TYPE, NEW DESTINATION, PRESSURE, SUSPICION. MIXED VALUES ARE SHOWN WITH A SLASH OR RANGE; *MARKS A LABEL THAT FOLLOWS ONE OF OUR CONVENTIONS.

Family	Items	Type	New	Press.	Susp.
invoice_routine	77/12/19	routine	N	N	0
notice	71/11/18	other	N	N	0
bec_polite	47/8/12	change	Y	N	2–3
change_legit	47/8/12	change	Y	N	1*
invoice_swap	29/5/7	routine	Y	N	2–3
fake_exec	47/8/12	exec	Y	Y	3
credit_note	47/8/12	credit	N	N	0
exec_legit	29/5/7	exec	N	Y*	1*
x402_overcharge	12/2/3	routine	N	N	2*
injection_redirect	24/4/6	change*	Y	Y/N	3
reminder_gentle	24/4/6	routine	N	N	0
x402_swap	29/5/7	change/routine	Y	Y/N	2–3
reminder_overdue	29/5/7	routine	N	Y	1*
x402_ok	29/5/7	routine	N	N	0
bec_urgent	35/6/9	change	Y	Y	3
refund_scam	12/2/3	credit*	Y	Y/N	3
injection_bypass	12/2/3	routine	N	Y	2

Metrics. Per question we report the accuracy of the top answer (a tie goes to the first option) and macro-F1. Calibration is the expected calibration error (ECE) [19], [20]: top label, 10 equal-width bins, pooled over the 4×150 answers (a stated probability on a bin edge counts in the upper bin). Routing is selective prediction [21] on p_{safe} . An item is *safe* to auto-clear when it is a routine invoice or credit note, keeps the registered destination and has suspicion at most 1 (51 test items); the other 99 should be *held*: 62 attacks and 37 legitimate items that still need a person (18 notices, 12 announced bank changes and 7 executive requests). Threshold-free, we report the AUROC of p_{safe} (safe against held items) and the most safe items one threshold clears while letting at most k held items through. At a 1% error budget (the lowest threshold whose cleared set is at most 1% held items) we report the share of safe items auto-cleared, *deployed* (threshold chosen on validation, applied to test) and *oracle* (chosen on test); with 150 items this budget tolerates no held item in the cleared set. We also report latency and cost per 1,000 items.

Protocol and fairness. Every model sees the same states and question strings but not our labelling rules: the fine-tunes learned them from 600 examples, the zero-shot models never saw them. Kev models are served through the same /v1/systemone API, in bf16 with MLX on one laptop. LLMs get one system prompt carrying every question’s instructions and every option’s criteria (the strings Kev receives), a short domain preamble and a JSON skeleton, and must return a probability per option. Llama 3.3 70B Instruct [22] (the FP8 variant on Cloudflare Workers AI) was called once per item at temperature 0; 0 of 150 replies failed to parse. The Claude models ran as Claude Code agents on a label-free copy of the test split with Llama’s prompt and a README noting that their probabilities drive the metrics: one agent per model, or three of 50 items for Haiku 4.5 after a run that wrote a keyword classifier was discarded. Each read about 5–10 items per step in one session, never revised an answer, and

could reason and run shell commands; Opus 5.5 used them for x402 unit conversions on five items, with amount ratios on all three overcharge items, and the single agents were told not to classify with code only as they finished. Their transcripts show no reads of labels, the repository or the web (paper/frontier-protocol.md). As a validity check, the kept runs give 111, 77, 77 and 72 distinct answers to the 150 items (Haiku, Sonnet, Opus, Fable), the discarded one 21. Temperatures and deployed thresholds come from validation; the zero-shot LLMs have no validation run, so only their oracle auto-clear is reported. Differences are paired over the same answers, with 95% intervals from an item-clustered bootstrap [23] (10,000 resamples, an item’s four answers kept together) and from a two-level bootstrap that resamples families, then items within them; p -values are exact two-sided sign tests over items. We call a difference significant when the item-level interval excludes zero and parity when it contains zero, and say when the family-level interval disagrees.

IV. METHOD

Our fine-tunes are supervised: cross-entropy on our labels, not reinforcement learning. payee-0.8b starts from Kev-0.8B [9] (jaredpalmer/kev-0.8b): a LoRA adapter [24] (rank 16, all target modules) and a pointer head that scores each option’s hidden state against the question’s, on a frozen Qwen3.5-0.8B-Base [25] (revision dc7cdfc); 11.3M parameters train. We continue from the released checkpoint (Hub snapshot 9a45d25, Kev trainer at commit f2bb629) for 2 epochs on the 600 training items: mean cross-entropy over the four questions, AdamW (weight decay 0.01), learning rate 2×10^{-5} on a one-cycle schedule, batch 1 with 8-step accumulation (150 optimizer steps), gradient-norm clipping at 1 (active on 148 steps), and Kev’s default augmentation (options reshuffled every epoch; “none of the above” replacing the answer or added as a wrong option, and an irrelevant option added, at 10%, 12% and 15%). We then fit one temperature on validation by minimum negative log-likelihood ($T=1.23$). Training took 39 minutes on an Apple M5 Max (48 GB; fp32 on MPS, each state encoded once for its four questions), 15 s per step at the median and at most 3.9 GB of GPU memory. We also fine-tuned Kev-4B (snapshot 139fdd9, 33.8M trainable parameters). The same recipe did not fit in 48 GB, so it used a bf16 frozen backbone, gradient checkpointing and one epoch (75 steps, 62 minutes, $T=1.11$); it was trained after the 0.8B’s test results were known. We also fine-tuned instruction-tuned open models on the same 600 items with LoRA in mlx-lm [26] (rank 8, learning rate 2×10^{-5} , effective batch 4, loss on the answer JSON only): Gemma 4 E2B [27] for 1 epoch (4 minutes) to fit the overnight window and Llama 3.2 3B [28] for 2 epochs (10 minutes). Each option is scored by its likelihood in that JSON, left to right. Qwen3-4B with the Kev recipe was not run: the overnight GPU window closed first (future work).

V. RESULTS

Table II gives the test results and Fig. 2 the paired differences. payee-0.8b reaches 0.918 mean accuracy: +17.2 points

over the released Kev-0.8B (95% CI [+14.2, +20.2]; family-clustered [+10.3, +24.3]; better on 91 items and worse on 6, sign test $p = 1.3 \times 10^{-20}$), and +12.3 over the five-times-larger released Kev-4B ([+9.3, +15.5]; [+5.0, +21.0]). Most of the gain is on suspicion, where the released models are near chance on the 0–3 scale (0.393 accuracy, mean error 1.09 levels) and the fine-tune reaches 0.787 (0.50 levels); the other three questions were already between 0.81 and 0.96 zero-shot. The gain is not mostly our conventions: without the convention answers it is +18.0 and +9.2 points. The fine-tuned 4B is statistically tied with the 0.8B in accuracy (the 0.8B’s lead is –2.0 points, CI [–4.2, +0.3], $p = 0.11$) but ranks safe against held items better (AUROC 0.986 against 0.944; difference CI [–0.078, –0.013], though the family-clustered interval [–0.169, +0.012] contains zero). Of the open models, our Gemma 4 E2B fine-tune is statistically tied with payee-0.8b in accuracy (payee-0.8b’s lead +2.0, CI [–1.2, +5.3]) but ranks safe against held items better (AUROC 0.990 against 0.944; difference CI [–0.084, –0.016], though the family-clustered interval [–0.175, +0.007] contains zero); our Llama 3.2 3B fine-tune is less accurate (+3.7, [+0.7, +6.7]; the family-clustered interval contains zero).

Against Llama 3.3 70B the lead is +10.3 points ([+6.7, +14.2]; family-clustered [+3.6, +19.3]), but it is a lead in following our labels, not in spotting fraud. On the 45 convention answers Llama scores 0.11 and payee-0.8b 0.62; without them the lead is +7.0 points ([+3.8, +10.3]). It sits on request type (+11.3) and the exact suspicion level (+22.7); new destination is tied (+2.0, [–2.0, +6.0]), and so is the binary split that p_{safe} uses, suspicion 0–1 against 2–3: Llama scores 0.893 against 0.833 (–6.0 points, [–12.7, +0.7]) whether the side is read from the top level or from the probability mass (an even split counting as held, as in routing). Llama’s clearest miss is invoices that hide an instruction to redirect payment: it flags the new destination on 2 and the suspicion level on 2 of the 6, against 6 and 6 for payee-0.8b. The fine-tune is weakest on silently swapped invoices (0.68): 4 of the 7 test items swap in a look-alike wallet that keeps the registered address’s first six and last four hex digits, which only a character-by-character comparison catches. Against the Claude models run as agents on the label-free kit, payee-0.8b’s lead is +5.5 over Haiku 4.5 ([+2.3, +8.8]), +1.0 over Sonnet 5 ([–1.8, +4.2]; tied), –3.5 over Opus 5.5 ([–5.8, –1.0]) and –3.2 over Fable 5.1 ([–5.5, –0.7]); the family-clustered intervals for Opus 5.5 and Fable 5.1 include zero. Our 4B fine-tune is statistically tied with Claude Sonnet 5, Opus 5.5 and Fable 5.1 (its lead +3.0, –1.5 and –1.2 points; every interval contains zero). On the 45 convention answers they score 0.31–0.58 (payee-0.8b 0.62), and on binary suspicion by top level 0.860–1.000 (payee-0.8b 0.833).

External check. We converted AgentDojo’s banking suite [12] into PayeeBench items: its 9 injection goals $\times$ 5 attack strings $\times$ 5 carrier documents (225 injected items) and 4 clean documents, labelled by our rules; and we added the 500 benign Japanese e-mails of PFN’s Japanese-Mail-Bench [29] (Table III; data in results/external). At its

TABLE II

PAYEEBENCH-JA TEST SPLIT (150 ITEMS): ACCURACY PER QUESTION, MEAN ACCURACY, ECE, AUROC OF p_{safe} , SAFE ITEMS AUTO-CLEARED AT A 1% ERROR BUDGET (DEPLOYED THRESHOLD, HELD ITEMS LET THROUGH, / ORACLE), p50 LATENCY AND COST PER 1,000 ITEMS (SECTION VII). [†]TEST SPLIT ONLY; LATENCY INCLUDES THE NETWORK. [‡]RUN AS AN AGENT ON THE LABEL-FREE KIT; COST ESTIMATED FROM LLAMA’S TOKEN COUNTS. —: NOT MEASURED.

Model	Req. type	New dest.	Pressure	Suspicion	Mean	ECE	AUROC	Auto-clear @1%	p50 (ms)	\$ / 1k
<i>Kev decision models, served locally</i>										
Kev-0.8B (released)	0.807	0.867	0.920	0.393	0.747	0.134	0.843	0% (0) / 12%	38	0.00013
Kev-0.8B + temp. refit on val.	0.807	0.867	0.920	0.393	0.747	0.106	0.844	0% (0) / 12%	38	0.00014
Kev-4B (released)	0.880	0.927	0.960	0.413	0.795	0.120	0.969	16% (0) / 33%	169	0.00060
payee-0.8b (ours)	0.947	0.967	0.973	0.787	0.918	0.024	0.944	45% (1) / 43%	39	0.00013
payee-4b (ours, 1 epoch)	0.953	0.987	1.000	0.813	0.938	0.016	0.986	67% (1) / 53%	165	0.00058
Qwen3-4B, Kev recipe (ours)	<i>not run: the overnight GPU window closed first (future work)</i>									
<i>Open generative models, LoRA SFT, local (ours)</i>										
Llama 3.2 3B + LoRA (ours)	0.887	0.993	0.933	0.713	0.882	0.033	0.916	27% (4) / 20%	–	–
Gemma 4 E2B + LoRA (ours)	0.880	0.987	0.920	0.807	0.898	0.050	0.990	61% (1) / 41%	–	–
<i>Zero-shot LLM</i>										
Llama 3.3 70B [†]	0.833	0.947	0.920	0.560	0.815	0.092	0.986	– / 39%	2,047	0.46
<i>Claude models run as Claude Code agents (indicative)</i>										
Claude Haiku 4.5 [‡]	0.887	0.933	0.947	0.687	0.863	0.035	0.932	– / 20%	–	1.3
Claude Sonnet 5 [‡]	0.867	1.000	0.940	0.827	0.908	0.097	1.000	– / 100%	–	2.5
Claude Opus 5.5 [‡]	0.993	1.000	0.940	0.880	0.953	0.067	1.000	– / 100%	–	5.1
Claude Fable 5.1 [‡]	0.993	1.000	0.940	0.867	0.950	0.081	1.000	– / 100%	–	13

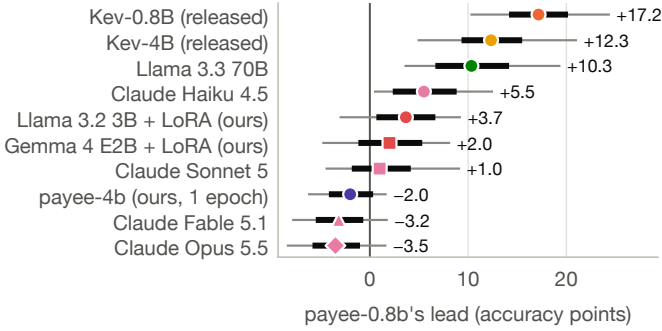


Fig. 2. payee-0.8b’s lead in mean test accuracy over each contender: point estimate, item-bootstrap 95% interval (thick) and family-clustered interval (thin). Negative values favour the contender.

deployed threshold payee-0.8b auto-clears 5 injected items (4 of them the password-change goal, which moves no money), and on the 180 in-scope items flags a new destination on 140 and suspicion 2 or more on 161, against 63 and 5 for the released 0.8B. It flags suspicion 2 or more on 8 benign e-mails (1.6%) and never a new destination. The items are English, Western and correlated (one attacker IBAN, fixed strings), only one clean document can be cleared, and the labels are our mapping, not AgentDojo’s attack-success metric.

VI. CALIBRATION AND ROUTING

The released Kev models are under-confident (Fig. 3): Kev-0.8B states 0.627 mean confidence at 0.747 accuracy and Kev-4B 0.708 at 0.795. Llama is over-confident: its 96 answers stated at about 0.80 were right 0.52 of the time (ECE 0.092). The Claude models’ ECE ranges from 0.035 (Haiku 4.5) to 0.097 (Sonnet 5), and they give 0–26 wrong answers at 0.9 confidence or more. payee-0.8b’s ECE of 0.024 is lower than every model’s except our 4B fine-tune’s (0.016). Pooled ECE

TABLE III

EXTERNAL CHECKS AT EACH MODEL’S DEPLOYED THRESHOLD: INJECTED ITEMS AUTO-CLEARED (OF 225), AND NEW-DESTINATION AND SUSPICION ≥ 2 FLAGS ON THE 180 IN-SCOPE ITEMS, THE 160 THAT MOVE MONEY AND 500 BENIGN JAPANESE E-MAILS.

Model	Cleared	In scope		Money moves		Benign mail	
		New dest.	Susp.	New / Susp.	New dest.	Susp.	
Kev-0.8B (released)	0	63	5	59 / 5	0	1	
Kev-4B (released)	0	152	157	152 / 152	0	0	
payee-0.8b (ours)	5	140	161	138 / 152	0	8	
payee-4b (ours, 1 epoch)	0	161	171	160 / 158	1	1	

hides confident errors, though: 13 of its answers are wrong at 0.9 confidence or more (released Kev-4B 0, Llama 22), and it rated 4 of the 6 benign gentle reminders in the test split (all four in Japanese) “very likely a scam” with probability 0.49 to 0.93.

Threshold-free, payee-0.8b ranks safe above held items with an AUROC of 0.944, better than the released Kev-0.8B (0.843) but not than the released Kev-4B (0.969; difference -0.025 , CI $[-0.062, +0.009]$) or Llama (0.986; -0.042 , $[-0.078, -0.014]$, though the family-clustered interval $[-0.164, +0.007]$ contains zero). With no held item cleared, or one, where a 1% budget operates, it clears 22 and 31 of the 51 safe items, against 17 and 22 for Kev-4B and 20 and 20 for Llama; these are differences of 2 to 11 items whose intervals contain zero (against Llama, $[-40.0, +35.3]$ and $[-46.0, +40.8]$ points of the safe items). Claude Sonnet 5, Opus 5.5 and Fable 5.1 rank every safe test item above every held one (AUROC 1.000), so a threshold chosen on test would clear all 51 safe items and none of the 99 others. Haiku 4.5 reaches 0.932 (clearing 10 with none held). With no validation run, their deployed rate is unknown.

With the threshold fixed on validation ($\tau = 0.884$), payee-0.8b clears 45% of the safe test items (23 of 51; 31% to

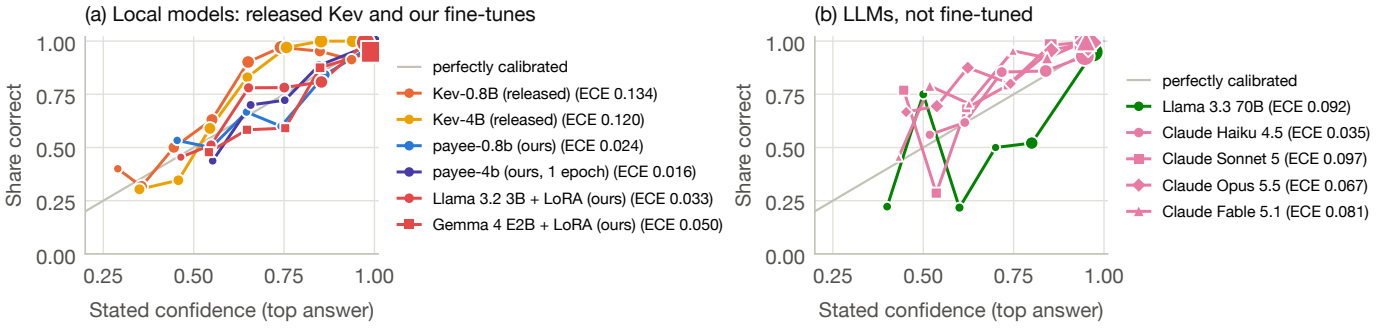


Fig. 3. Reliability on the test split: share of answers correct against stated confidence, in 10 bins over the four questions pooled. Marker area grows with the answers in a bin; bins with fewer than 10 are omitted.

59% when test items are resampled) and lets 1 held item through: an invoice whose 振込先 moved to another bank under the same account name, which the kernel’s registry match exists to catch. The fine-tuned 4B clears 67%, also with 1 held item; the released Kev-4B 16%, and the released 0.8B nothing, even with a refitted temperature. This is not like-for-like: validation shares training templates and fitted the temperature, so the fine-tunes rank it almost perfectly (AUROC 0.989 for payee-0.8b) and their threshold is loose on test, while the released Kev-4B ranks validation worse than test (0.937 against 0.969). A production threshold should come from real held-out documents, with a margin.

VII. COST AND SPEED

Latency is the client’s wall time for one HTTP request answering all four questions, one request at a time, after warm-up. payee-0.8b answers in 39 ms at p50 (44 ms at p95), the same as the released 0.8B; the 4B models take about 165 ms. These are over localhost; Llama’s 2,047 ms (5,616 ms at p95) includes the network and our proxy. A local model has no token bill, so we price the time a request occupies the machine, $t_{\text{item}} \cdot W \cdot c_{\text{kWh}}$, with t_{item} the mean wall time per item, $W = 60$ W an assumed package power and $c_{\text{kWh}} = \$0.21$ per kWh (a Tokyo household tariff), excluding the hardware: \$0.00013 per 1,000 items for payee-0.8b. Hosted models are priced at list price for their measured tokens: Llama 3.3 70B costs \$0.46 per 1,000 items [30]. At list price for the token counts Llama used on the same prompts, the Claude models would cost \$1.3 (Haiku 4.5) to \$13 (Fable 5.1) per 1,000 items [31], probably an underestimate: their billed reasoning tokens are not counted, and their latency was not measured. Llama costs about 3,400 times our electricity, but electricity on owned hardware against a list price is not a like-for-like ratio. On a rented NVIDIA L4 at \$0.80 per hour and Kev’s published throughput of 62.8 requests per second (a different workload: short texts, many concurrent clients), Kev-0.8B would cost about \$0.0035 per 1,000 items, still about 129 times less than Llama.

VIII. FORGETTING

On Kev’s own out-of-domain development suite (transfer-v4, 656 answers, served the same way) accuracy held: 0.651

to 0.637 for the 0.8B and 0.817 to 0.817 for the 4B. Calibration did not: ECE rose from 0.049 to 0.225 (0.8B) and from 0.033 to 0.131 (4B), and wrong answers given with at least 0.9 confidence rose from 0.2% to 12.8% (0.8B). The fitted temperature belongs to the payee questions; Meigi serves the fine-tune for this question set only.

IX. LIMITATIONS

PayeeBench-JA is synthetic and self-built: it measures how well a model learns our families and conventions, not fraud detection in real mail with OCR noise, threads and attachments. The test set is small and clustered: intervals are several points wide, family-clustered ones about twice as wide, and one or two items decide the 1% auto-clear figure. The test split was reused: the models were trained on an earlier build that differs only in identifiers, every model was scored once per build, the 4B was trained after the 0.8B’s results were known and Gemma 4 E2B after Llama 3.2 3B’s, and the 0.8B was chosen on test. Each model was trained once, with one seed; a first Llama 3.2 3B run at a higher learning rate diverged and was stopped unscored. The frontier rows are agents with context, reasoning and a shell, run once, with no latency and an estimated cost. TypeSafe’s hosted System-1 model, Jev [8], was not evaluated.

Audit. An AI auditing agent re-derived the headline numbers with independent code and found one bug, fixed throughout: a probability exactly on a bin edge fell into the lower bin, understating Llama’s ECE (0.076, now 0.092). Its other findings narrowed our claims and made the agent rows indicative. A second AI review recomputed the headline numbers, frontier rows included, and a third the open-model rows.

X. CONCLUSION

On PayeeBench-JA, a 0.8B decision model fine-tuned on 600 synthetic items in 39 minutes on a laptop learns our labels: it is +17.2 points more accurate than the released model, its calibration error is lower than every model’s except our 4B fine-tune’s (0.016), and it answers in 39 ms at a negligible marginal cost. It is not a better detector of redirection than large LLMs, which rank safe against held items at least as well, and a Gemma 4 E2B fine-tuned on the same items is

statistically tied with it in accuracy and ranks safe against held items better (item-level; the family-clustered interval includes zero). Run as agents, Claude Opus 5.5 and Fable 5.1 are also more accurate than it (item-level; the family-clustered intervals include zero) and three Claude models separate safe from held items perfectly. What it offers a payment agent is a local, cheap first pass with mostly calibrated probabilities that never needs to be trusted with money: the model only routes, and money moves only through exact, on-chain checks.

REPRODUCIBILITY

Everything is under bench/ in the Meigi repository: the dataset builder (byte-identical output given the national corporation index, which is not in the repository), the fine-tuning and evaluation scripts, the label-free kit and its scorer, the agents' instructions, and the paper/ scripts that generate every number, table and figure from results/ and dataset/. The fine-tuned weights are not published: their predictions can be re-scored, not re-run.

ACKNOWLEDGMENT

This paper was written with AI assistance (Claude) and audited by AI agents. All numbers come from bench/results and bench/dataset, through the scripts above.

REFERENCES

- [1] Federal Bureau of Investigation, Internet Crime Complaint Center, "Internet crime report 2024," 2025, accessed 2026-09-26. [Online]. Available: https://www.ic3.gov/AnnualReport/Reports/2024_IC3Report.pdf
- [2] Kyodo, "Japan Airlines falls victim to email fraud, paying out ¥384 million to Hong Kong accounts," The Japan Times, 21 Dec. 2017, accessed 2026-09-26. [Online]. Available: <https://www.japantimes.co.jp/news/2017/12/21/business/japan-airlines-bilked-%C2%A5384-million-getting-bogus-emails-seeking-lease-fees/>
- [3] NHK, "日本航空偽メールにだまされ 3 億 8000 万円余被害 [Japan Airlines loses over ¥380 million to fake e-mails]," NHK News Web, 20 Dec. 2017, accessed 2026-09-26. [Online]. Available: <https://www3.nhk.or.jp/news/html/20171220/k10011265791000.html>
- [4] Pay.UK, "Confirmation of payee," accessed 2026-09-26. [Online]. Available: <https://www.wearepay.uk/what-we-do/overlay-services/confirmation-of-payee/>
- [5] European Parliament and Council of the European Union, "Regulation (EU) 2024/886 of 13 March 2024 as regards instant credit transfers in euro," Official Journal of the European Union, L series, 19 Mar. 2024, accessed 2026-09-26. [Online]. Available: <https://eur-lex.europa.eu/eli/reg/2024/886/oj>
- [6] x402 Foundation, "x402: A payments protocol for the internet, built on HTTP," 2025, originally coinbase/x402. Accessed 2026-09-26. [Online]. Available: <https://github.com/x402-foundation/x402>
- [7] National Tax Agency, Japan, "インボイス制度について [the qualified invoice system]," accessed 2026-09-26. [Online]. Available: https://www.nta.go.jp/taxes/shiraberu/zeimokubetsu/shohi/keigenzeiritsu/invoice_about.htm
- [8] TypeSafe AI, "Models," accessed 2026-09-26. [Online]. Available: <https://docs.typesafe.ai/models.md>
- [9] J. Palmer, "Kev: Small Jev-like decision models you can train and run yourself," 2026, Apache-2.0 licence. Accessed 2026-09-26. [Online]. Available: <https://github.com/jaredpalmer/kev>
- [10] A. I. Champa, M. F. Rabbi, and M. F. Zibran, "Curated datasets and feature analysis for phishing email detection with machine learning," in *2024 IEEE 3rd International Conference on Computing and Machine Intelligence (ICMI)*. IEEE, 2024, pp. 1–7.
- [11] A. El Aassal, S. Baki, A. Das, and R. M. Verma, "An in-depth benchmarking and evaluation of phishing detection research for security needs," *IEEE Access*, vol. 8, pp. 22 170–22 192, 2020.
- [12] E. DeBenedetti, J. Zhang, M. Balunovic, L. Beurer-Kellner, M. Fischer, and F. Tramèr, "AgentDojo: A dynamic environment to evaluate prompt injection attacks and defenses for LLM agents," in *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, vol. 37, 2024, pp. 82 895–82 920.
- [13] Q. Zhan, Z. Liang, Z. Ying, and D. Kang, "InjecAgent: Benchmarking indirect prompt injections in tool-integrated large language model agents," in *Findings of the Association for Computational Linguistics: ACL 2024*, 2024, pp. 10 471–10 506.
- [14] A. S. Patlan, P. Sheng, S. A. Hebbbar, P. Mittal, and P. Viswanath, "Real AI agents with fake memories: Fatal context manipulation attacks on Web3 agents," arXiv:2503.16248, 2025. [Online]. Available: <https://arxiv.org/abs/2503.16248>
- [15] U. Uchibeke, "APort Vault: Benchmarking AI agent payment authorization with the Open Agent Passport," arXiv:2609.22076, 2026. [Online]. Available: <https://arxiv.org/abs/2609.22076>
- [16] Y. Louck, "Protocol-level attacks on agentic commerce platforms: A cross-platform taxonomy, AIP-Bench, and unified defense," arXiv:2607.21824, 2026. [Online]. Available: <https://arxiv.org/abs/2607.21824>
- [17] H. Gong, M. Samawi, and F. Medda, "Authority-inference separation in agentic finance: First-line control, blockchain enforcement, and replayable assurance," arXiv:2608.30519, 2026. [Online]. Available: <https://arxiv.org/abs/2608.30519>
- [18] National Tax Agency, Japan, "チェックデジットの計算 [calculation of the corporate-number check digit]," accessed 2026-09-26. [Online]. Available: <https://www.houjin-bangou.nta.go.jp/documents/checkdigit.pdf>
- [19] M. P. Naeini, G. F. Cooper, and M. Hauskrecht, "Obtaining well calibrated probabilities using Bayesian binning," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 29, no. 1, 2015, pp. 2901–2907.
- [20] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proceedings of the 34th International Conference on Machine Learning (ICML)*, ser. PMLR, vol. 70, 2017, pp. 1321–1330.
- [21] Y. Geifman and R. El-Yaniv, "Selective classification for deep neural networks," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017.
- [22] Meta, "Llama-3.3-70B-Instruct model card," 2024, accessed 2026-09-26. [Online]. Available: <https://huggingface.co/meta-llama/Llama-3.3-70B-Instruct>
- [23] B. Efron, "Bootstrap methods: Another look at the jackknife," *The Annals of Statistics*, vol. 7, no. 1, pp. 1–26, 1979.
- [24] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "LoRA: Low-rank adaptation of large language models," in *International Conference on Learning Representations (ICLR)*, 2022.
- [25] Qwen Team, "Qwen3.5-0.8B-Base," 2026, accessed 2026-09-26. [Online]. Available: <https://huggingface.co/Qwen/Qwen3.5-0.8B-Base>
- [26] MLX Contributors, "MLX LM," Python package mlx-lm, version 0.31.3, 2026, accessed 2026-09-26. [Online]. Available: <https://github.com/ml-explore/mlx-lm>
- [27] Google, "gemma-4-E2B-it," Hugging Face model google/gemma-4-E2B-it, revision 3e22461, 2026, Apache-2.0 licence. Accessed 2026-09-26. [Online]. Available: <https://huggingface.co/google/gemma-4-E2B-it>
- [28] Meta, "Llama-3.2-3B-Instruct," Hugging Face model; we used the MLX conversion mlx-community/Llama-3.2-3B-Instruct, revision 96e4e2c, 2024, accessed 2026-09-26. [Online]. Available: <https://huggingface.co/meta-llama/Llama-3.2-3B-Instruct>
- [29] Preferred Networks, "Japanese-Mail-Bench: A Japanese business e-mail generation benchmark," Hugging Face dataset pfnet/japanese-mail-bench, revision 4ed1a7d, 2026, Apache-2.0 licence. Accessed 2026-09-26. [Online]. Available: <https://huggingface.co/datasets/pfnet/japanese-mail-bench>
- [30] Cloudflare, "Workers AI pricing," accessed 2026-09-26. [Online]. Available: <https://developers.cloudflare.com/workers-ai/platform/pricing/>
- [31] Anthropic, "Claude API pricing," accessed 2026-09-26. [Online]. Available: <https://platform.claude.com/docs/en/about-claude/pricing>